**How otters stay warm**
The smallest mammal in the ocean doesn't rely on blubber or a large body to keep toasty. **https://dgit.in/jul21-89**

**mRNA based flu shot**
The first group of participants in a clinical study received Moderna's mRNA-based seasonal flu shot. **https://dgit.in/jul21-91**

Benjamin Libet. You can google him and his studies if you want to know more about them. The gist of it was that the experiments he and his colleagues ran found that there was a buildup of brainwaves just before a conscious event happened (like someone pressing a button). And this buildup was happening even before the person was aware of their decision to act. This buildup has been called the "readiness potential" of the brain, and we seem to be totally unaware of this happening.

It's no wonder that this study has been used by many to declare that there is no such thing as free will. If our conscious mind thinks we have made up our mind to do something at a point Y in time, and if EEG readings show that the brain itself is gearing up for the action anywhere from a few hundred milliseconds to as much as a free second before Y, then are we even in control? If our brains know we are going to act several seconds before we ourselves do, how can we claim to be in control? Maybe what we think of as control is merely our brain informing us of the decision it has already taken, just before acting.

## DISAGREEMENT

The side of the scientific community who is on the other side of the argument claims that the readiness potential has not been proven to be anything other than the brain gearing up to make a decision. If you were a sprinter at any point in your life, you know that you are supposed to start running only when you hear the starter's pistol or the word "Go" if using that as a starting phrase. We all know that "Go" is preceded by "On your Marks", and also, "Get set". And do our bodies tense up when we hear "Get set"? Of course! All our running muscles tense, our brain is hyper focussed waiting for the signal that indicates that we can start running, and a few of us let the anticipation get the best of us and start running prematurely, resulting in a false start. So could the readiness potential be just that? The brain gearing up to make a decision as the conscious mind decides when to act? In order to do away with delays in actions, our brain might be fine-tuned to prepare itself for a decision – and decisions might have an "On your marks", and "Get set" aspect to them that we just aren't aware of.

Others point out that Libet's experiments (and all experiments like his), rely on a self-timing of sorts by the test subjects, which is in itself a flawed methodology. While it's true that there's no way we can know when someone's brain made a conscious decision, it's also true that making decisions doesn't usually involve an additional layer of trying to note when it is you made that decision... you don't usually sit around thinking "Ok press the button NOW!", while also trying to remember the exact time on a clock when you decided "NOW!". This introduces delays in the system which can explain the delay between the readiness potential signal's rise and the peak signal of the brain initiating the movement. Open the stopwatch app on your phone, let it run, stare at it and try and find the exact time you consciously twitch a finger. Not easy is it?

Other studies which involve seeing images and pressing one of two buttons (such as press Button A if the image is black and white, or Button B if it is in colour) shows the same readiness potential signal in the brain even before the image is shown! If that's the case, how can the readiness potential be making a decision, because surely we cannot be deciding on which button to press even before the experiment (of showing the image) has happened?

Philosophers and other scientists suggest that perhaps it is wrong to even separate the subconscious brain from the conscious one. After all, our brains are efficient enough to learn something and then not need to re-learn or re-think everything to play out every aspect of every decision in our conscious mind.

We all have muscle memory, and knee-jerk reactions to things which we have already encountered before. It's how so many of us are fooled by magic and sleight of hand, and illusions. It's also how we are able to function efficiently without going mad. A simple example is that this writer uses pattern matching to play the game Minesweeper. Most people never progress past playing it a few times, because it can be quite an irritating game to understand. When you open a square, the number tells you how many squares (closed or opened) touching that square have mines placed on them. Figuring out that for every opened square can be quite painful, as these numbers can range from nothing (0)


Pattern matching used to mark the mines

to 8. However, most players who stick to playing the game and start playing it very fast are never figuring out individual square. Instead, they've just memorised the patterns of numbers and apply those few sets patterns to the grids in front of them. For example, a pattern of 1-2-1 along a wall of closed squares means that the squares directly below the 1s are mines, and the square below the 2 is not!

So maybe our brains just function that way, and instead of the conscious brain being involved in everything, maybe the whole brain works as one, with only brand-new, never-thought-of-before choices being presented to the conscious mind. We need to study this in much more detail before science can ever arrive at the answer of whether free will exists. For now, both sides of the argument claim to be right.